

Q1)

```
[→ isc git:(main) ./hw2q1
Enter a positive integer: 6173
6173 18520 9260 4630 2315 6946 3473 10420 5210 2605 7816 3908 1954 977 2932 1466 733 2200 1100 550 275 826 413 1240 620
310 155 466 233 700 350 175 526 263 790 395 1186 593 1780 890 445 1336 668 334 167 502 251 754 377 1132 566 283 850 425
1276 638 319 958 479 1438 719 2158 1079 3238 1619 4858 2429 7288 3644 1822 911 2734 1367 4102 2051 6154 3077 9232 4616 2
308 1154 577 1732 866 433 1300 650 325 976 488 244 122 61 184 92 46 23 70 35 106 53 160 80 40 20 10 5 16 8 4 2 1
Total shopping time: 111
Max value reached: 18520
→ isc git:(main) █
```

Here is the Collatz conjecture on 6,173. It demonstrates the total shopping time, defined by the number of iterations it takes to reach 1. It also shows the highest value reached along getting to 1.

Q2)

```
→ isc git:(main) ./hw2q2
Enter a positive integer: 987654321
Enter another positive integer: 9999
gcd(987654321,9999) = 9
Number of iterations: 9
[→ isc git:(main) ./hw2q2
Enter a positive integer: 9999
Enter another positive integer: 987654321
gcd(9999,987654321) = 9
Number of iterations: 9
→ isc git:(main) █
```

Here is the demonstration of Euclid's algorithm in effect. Using both 987654321, 9999 as well as 9999,987654321 demonstrates the programs ability to function regardless of whether the first or second input is greater. The program shows the number of iterations of the loop it takes to solve the gcd as well as the gcd.